

GPIE

Green Policy Intelligence Engine

All Maps and Plots

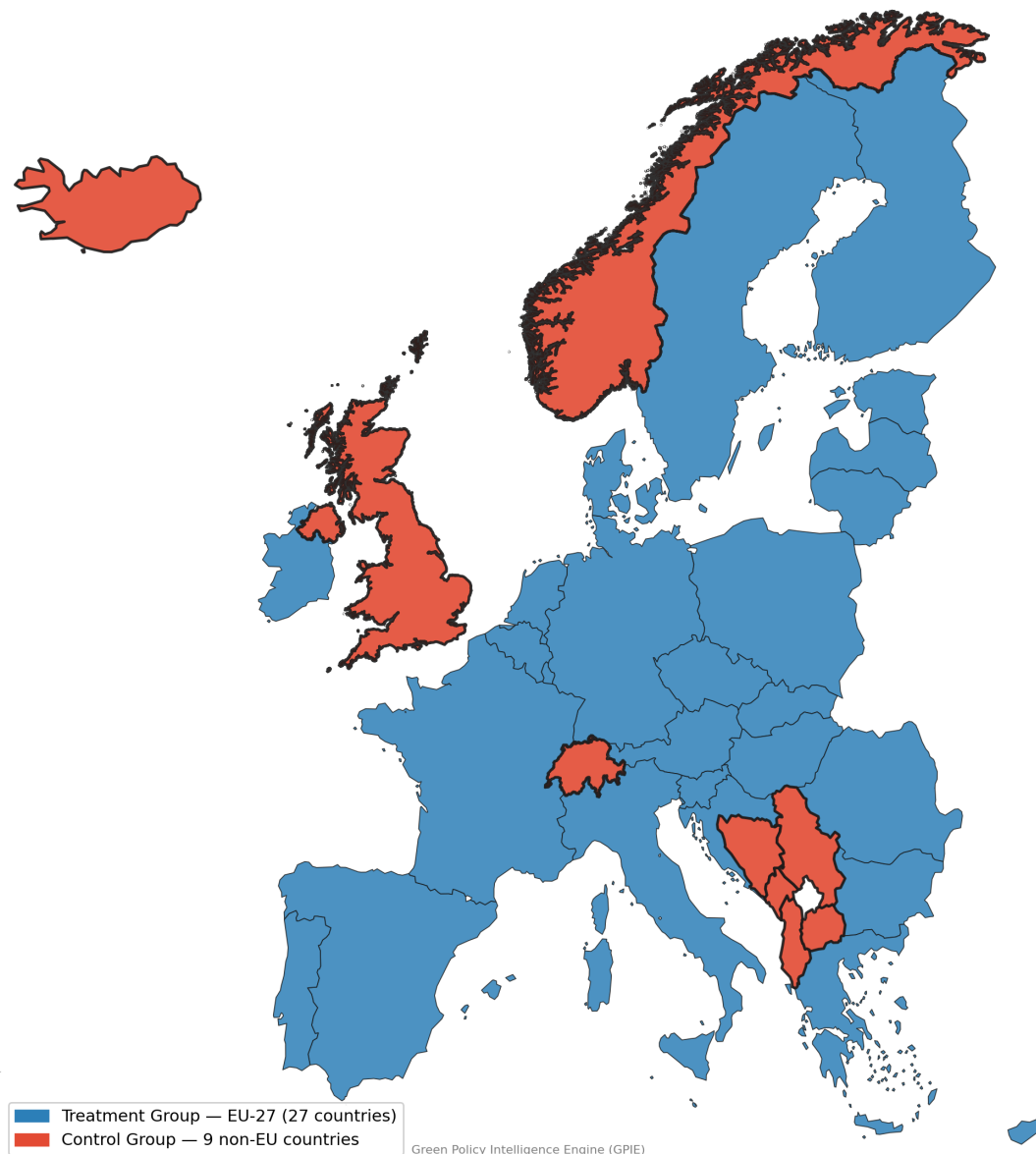
20 figures compiled from outputs/plots/

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Study Design: Treatment vs. Control Group

EU-27 (treatment) and the 9-country non-EU control group - UK, Norway, Switzerland, Iceland, Albania, Bosnia and Herzegovina, Montenegro, North Macedonia, Serbia.

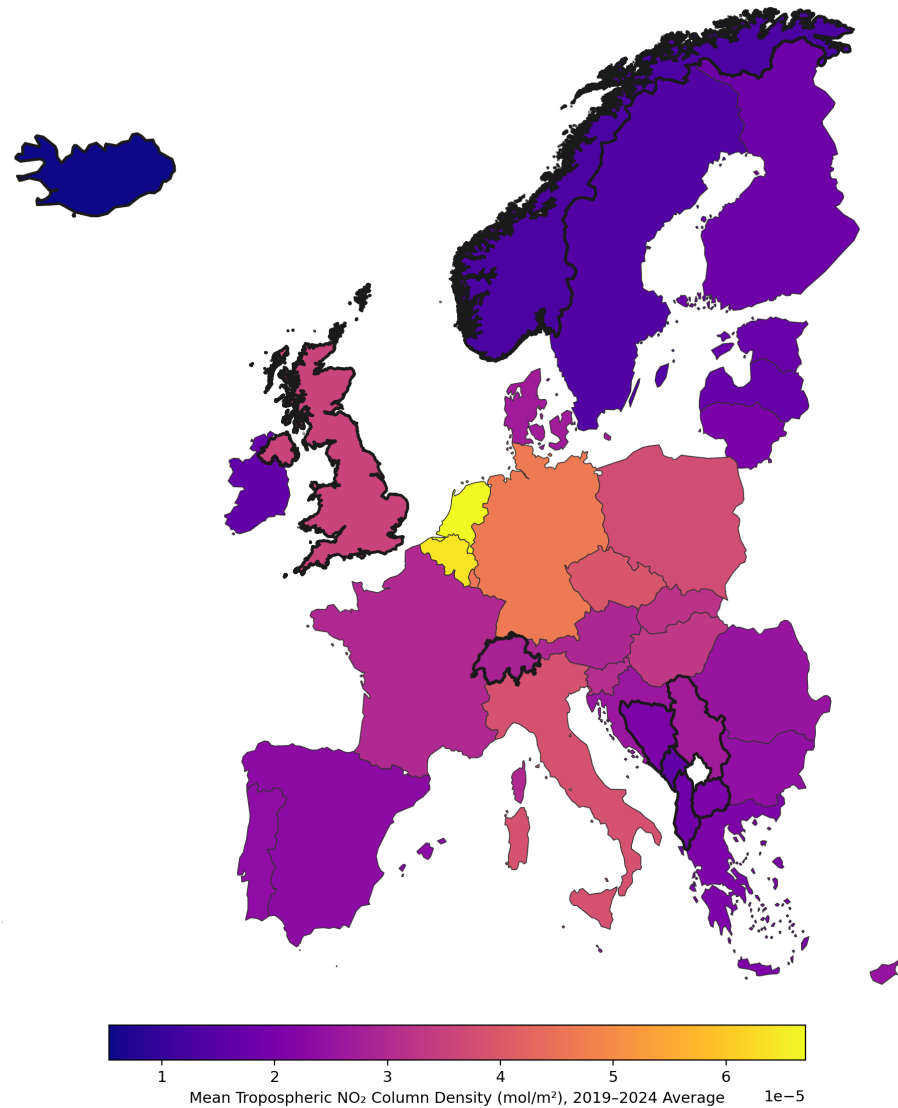
GPIE Study Design: Treatment vs. Control Group
Difference-in-Differences comparison — EU-27 (subject to the European Climate Law) vs. nine non-EU comparator countries
UK, Norway, Switzerland, Iceland, Albania, Bosnia and Herzegovina, Montenegro, North Macedonia, Serbia



NO2 Choropleth Map

Mean tropospheric NO₂ concentration by country, Sentinel-5P TROPOMI.

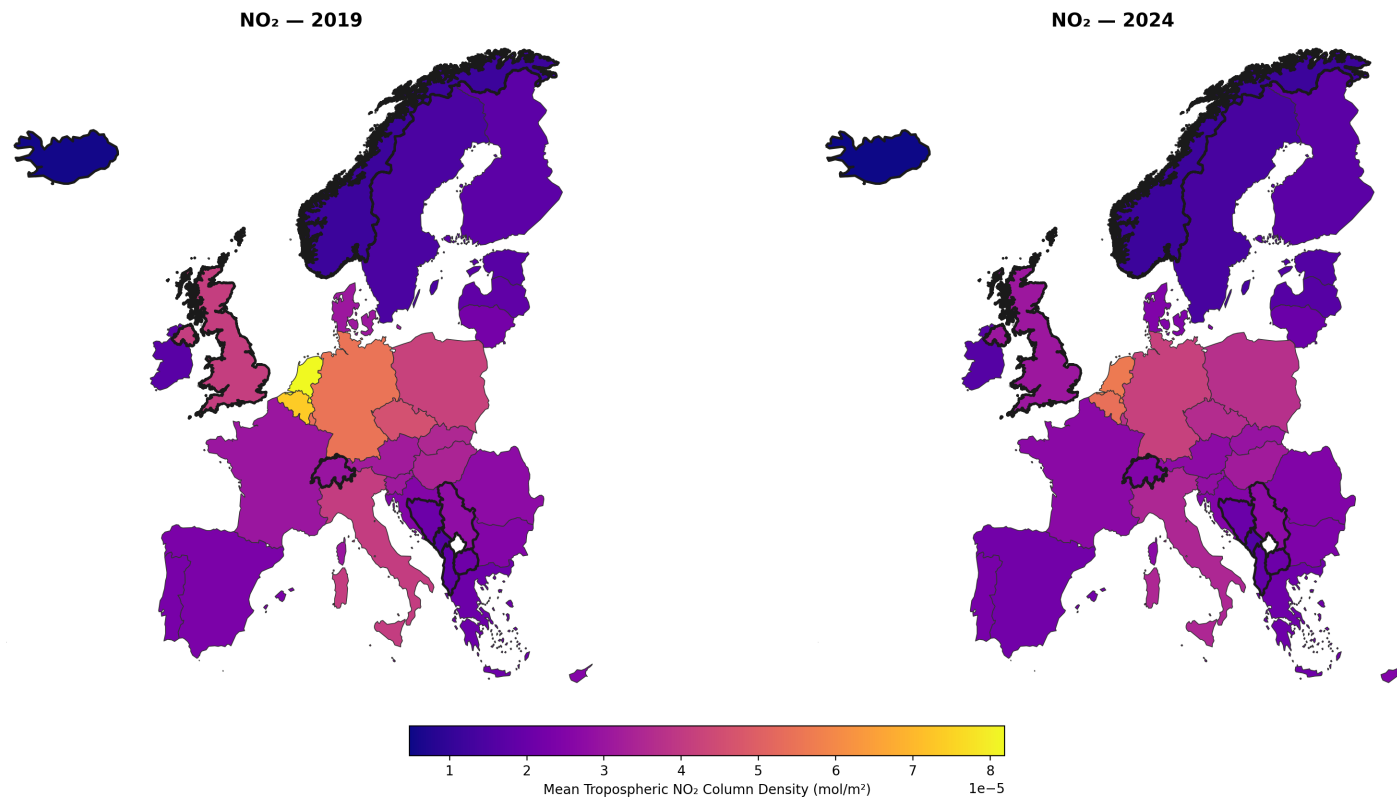
Tropospheric NO₂ Column Density: EU-27 vs. 9-Country Control Group
Thick borders mark non-EU control-group countries — descriptive map, 2019–2024 mean (mol/m²)



NO₂ Before vs. After the European Climate Law

Country-level annual mean NO₂, all 36 countries, 2019 (pre-treatment) vs. 2024 (post-treatment). Descriptive only.

NO₂ Levels Before vs. After the European Climate Law: EU-27 and Control Group (2019 vs. 2024)
Thick borders mark the 9-country non-EU control group — descriptive comparison; see the DiD model for inference

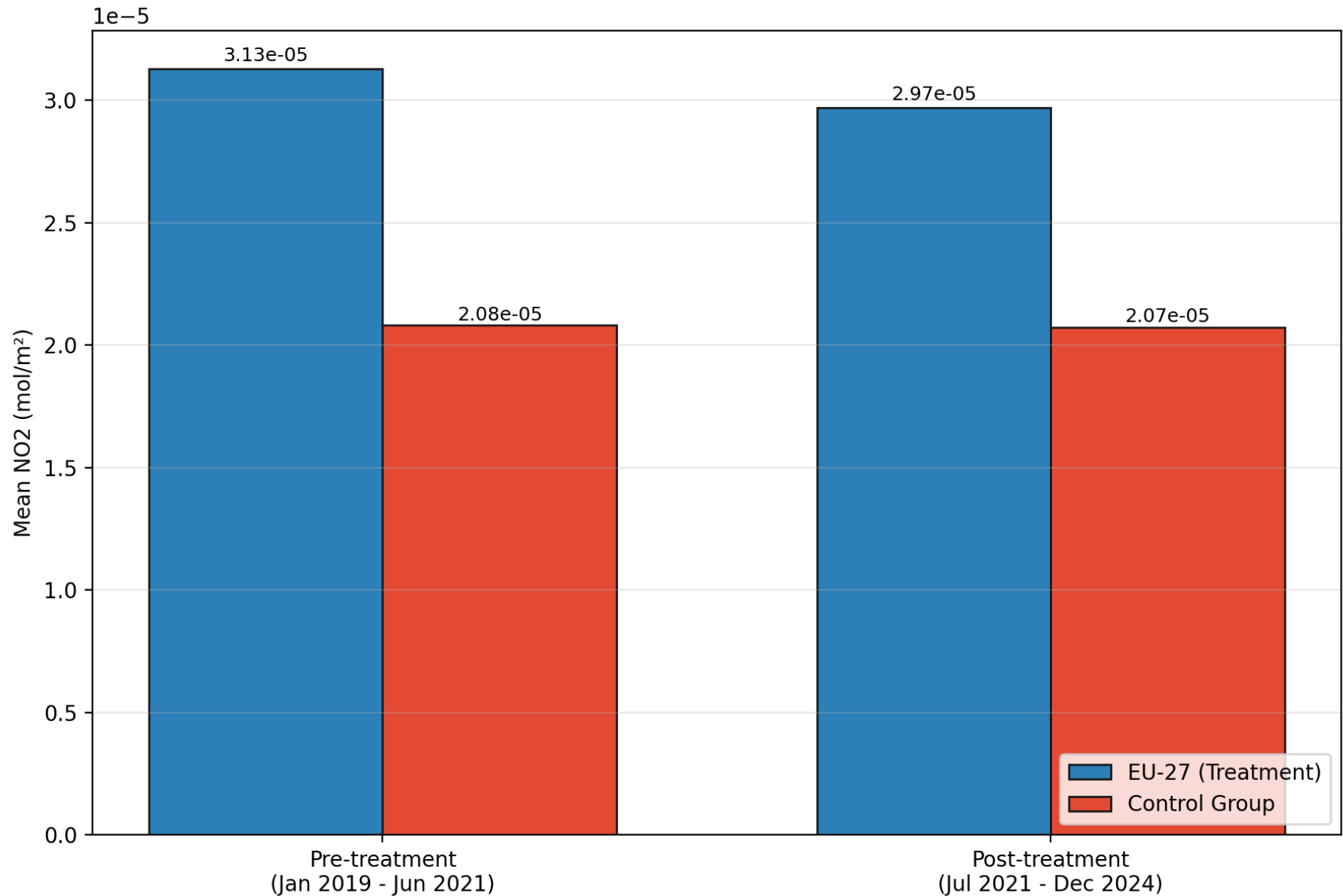


Green Policy Intelligence Engine (GPiE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Process API

NO2: EU-27 vs. Control Group

Raw mean NO2, EU-27 vs. 9-country control group, Jan 2019-Jun 2021 vs. Jul 2021-Dec 2024 (descriptive; DiD estimate in title).

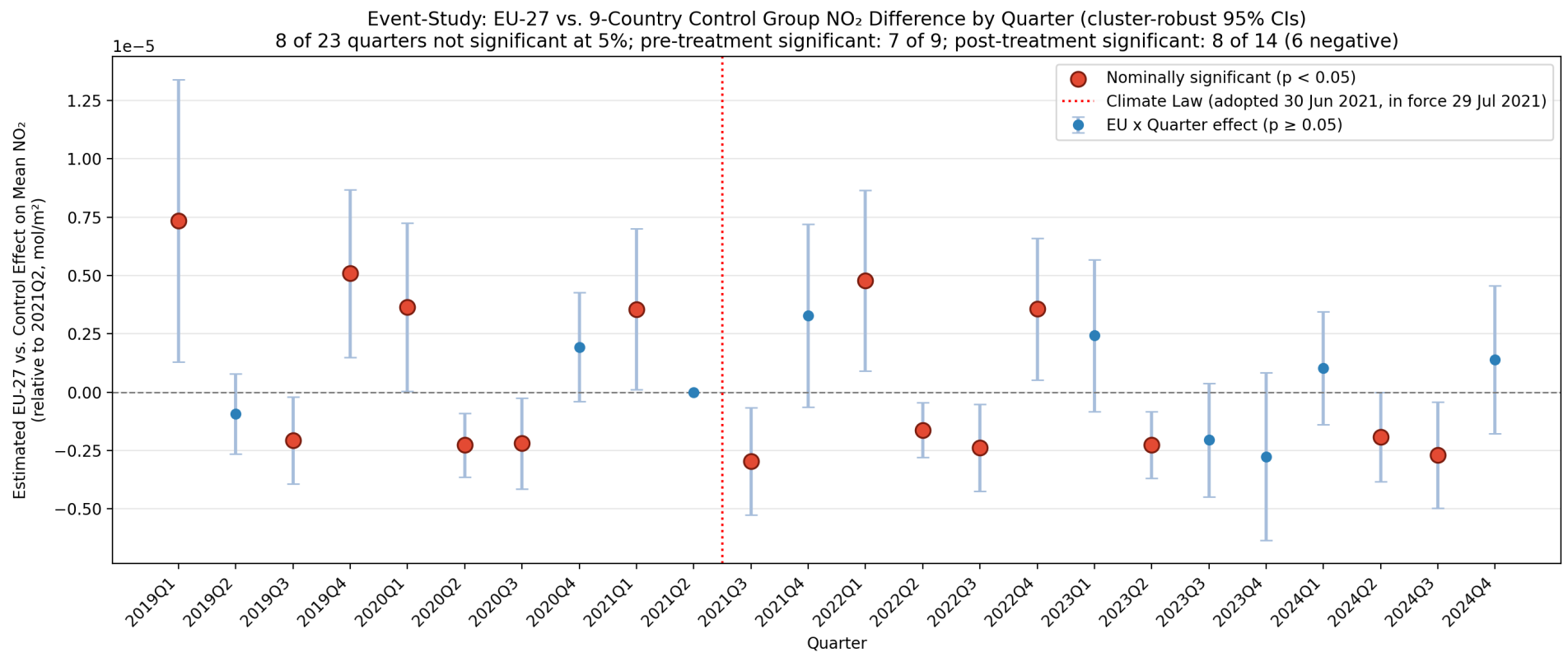
NO2: EU-27 vs. 9-Country Control Group, Before vs. After the European Climate Law
Raw group means (descriptive). Two-group DiD estimate: $-1.52\text{e-}06$, $p = 0.013$ (cluster-robust)



Green Policy Intelligence Engine (GPIE) - Source: Sentinel-5P TROPOMI, Sentinel Hub Process API

Event-Study: Quarter-by-Quarter NO₂ Effect

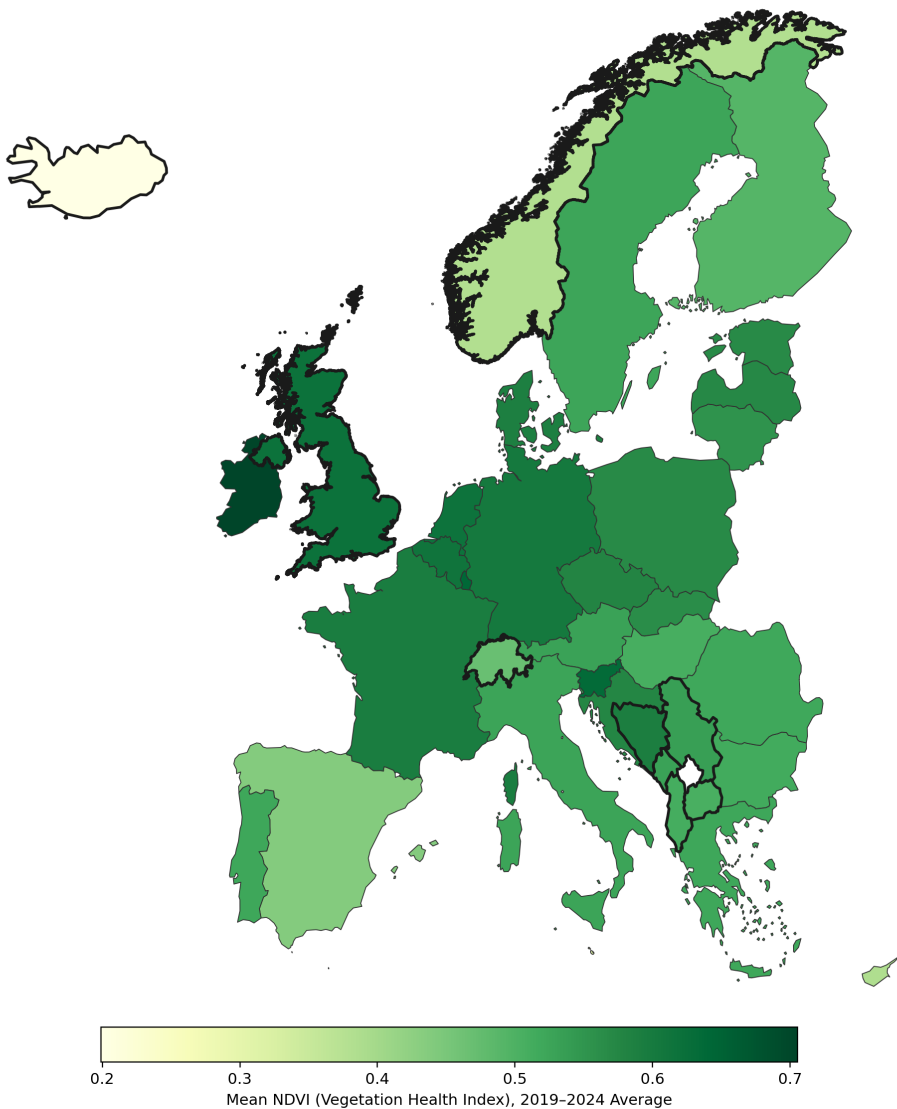
23 quarter-specific EU x quarter coefficients relative to 2021Q2, cluster-robust 95% CIs (counts of significant quarters in the figure title).



NDVI Choropleth Map

Mean vegetation health index (NDVI) by country.

Vegetation Health (NDVI): EU-27 vs. 9-Country Control Group
Thick borders mark non-EU control-group countries



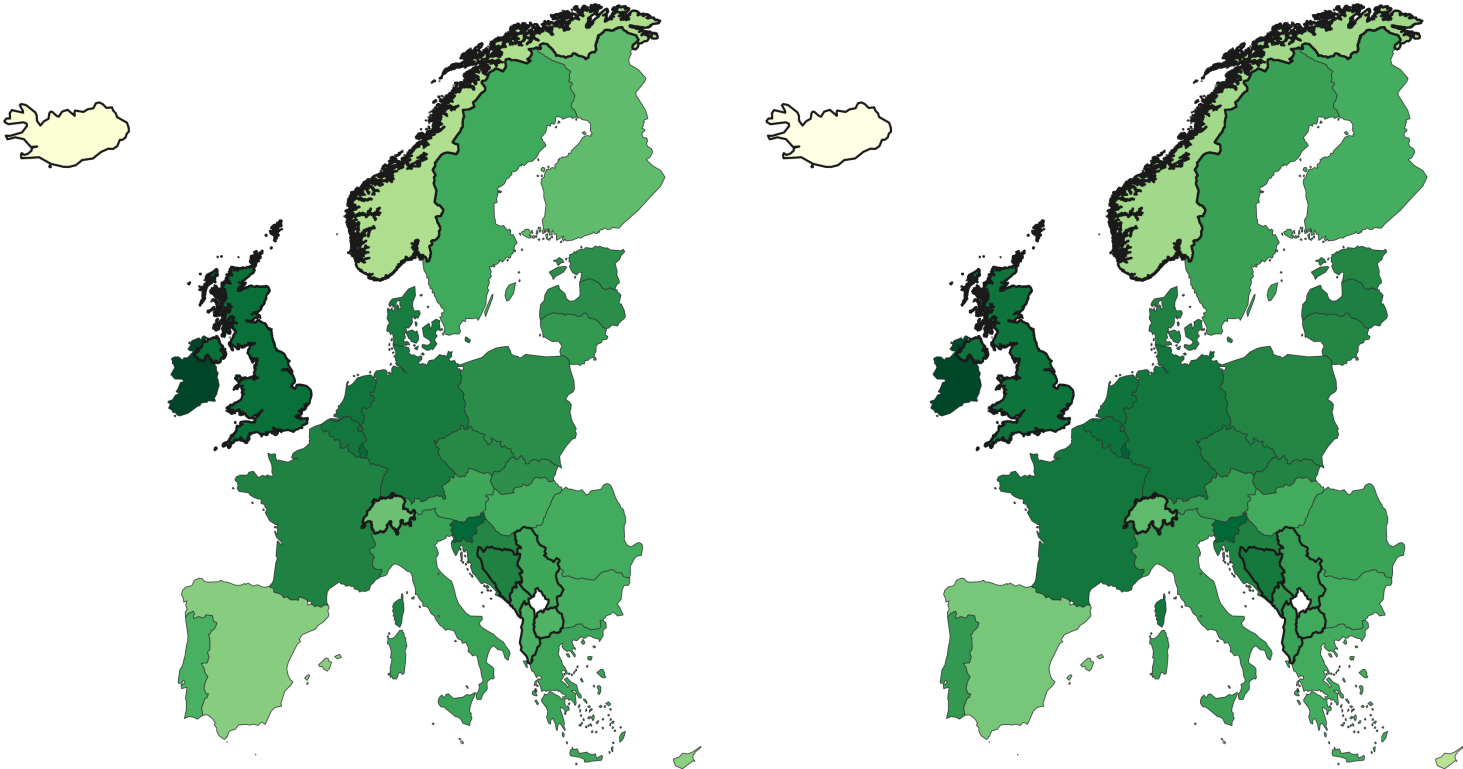
NDVI Before vs. After the European Climate Law

Country-level annual mean NDVI, all 36 countries, 2019 vs. 2024. Descriptive only.

NDVI (Vegetation Health) Before vs. After the European Climate Law: EU-27 and Control Group (2019 vs. 2024)
Thick borders mark the 9-country non-EU control group

NDVI — 2019

NDVI — 2024

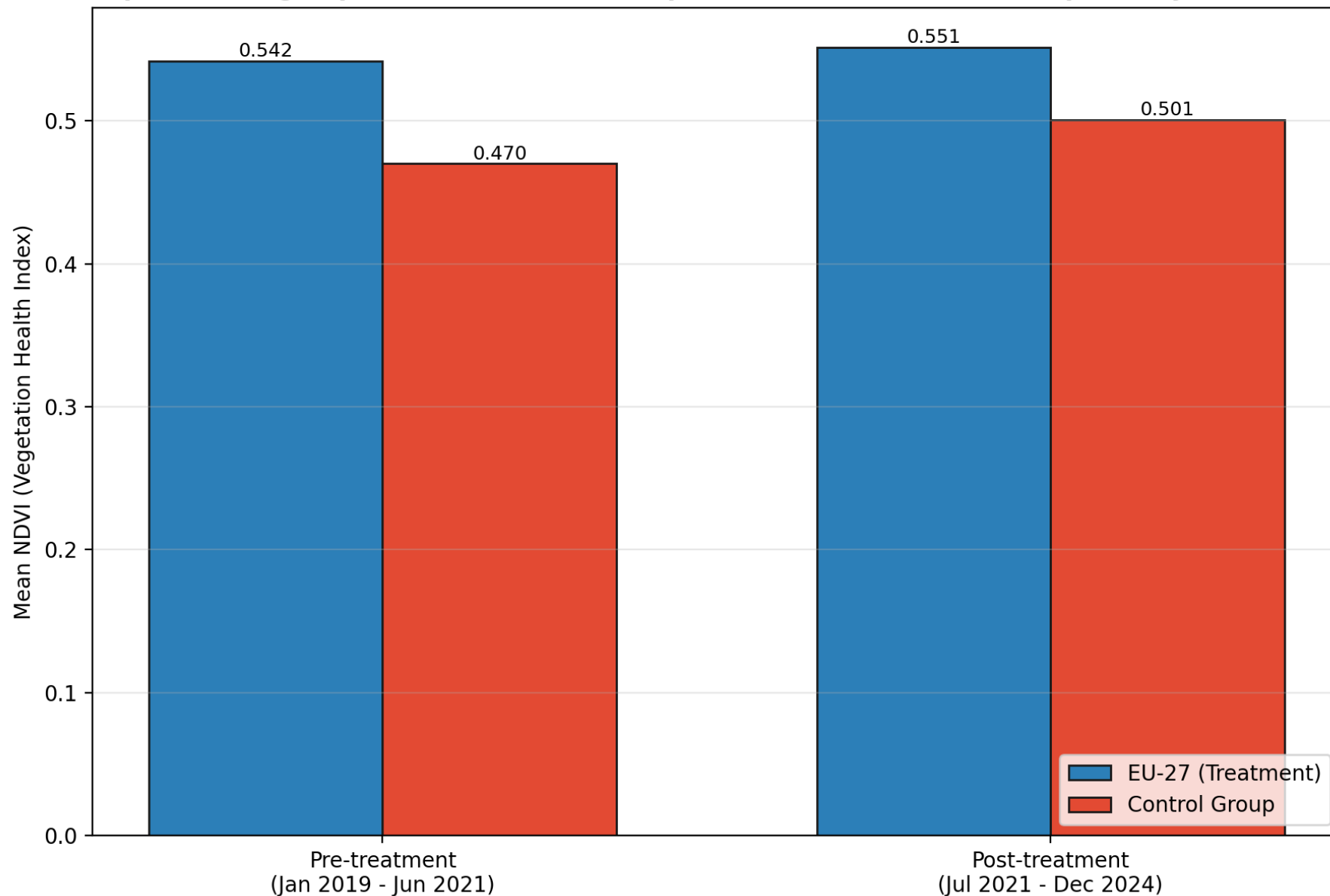


Green Policy Intelligence Engine (GPIE) — Source: CGLS NDVI 300m, Sentinel Hub Statistical API

NDVI: EU-27 vs. Control Group

Raw mean NDVI, EU-27 vs. control group, pre vs. post (descriptive); exploratory DiD estimate in title, not attributed to the Climate Law.

**NDVI: EU-27 vs. 9-Country Control Group, Before vs. After the Climate Law date
ins (descriptive). Two-group DiD estimate: -0.0194, p = 0.005 (cluster-robust) - exploratory, not attribute**



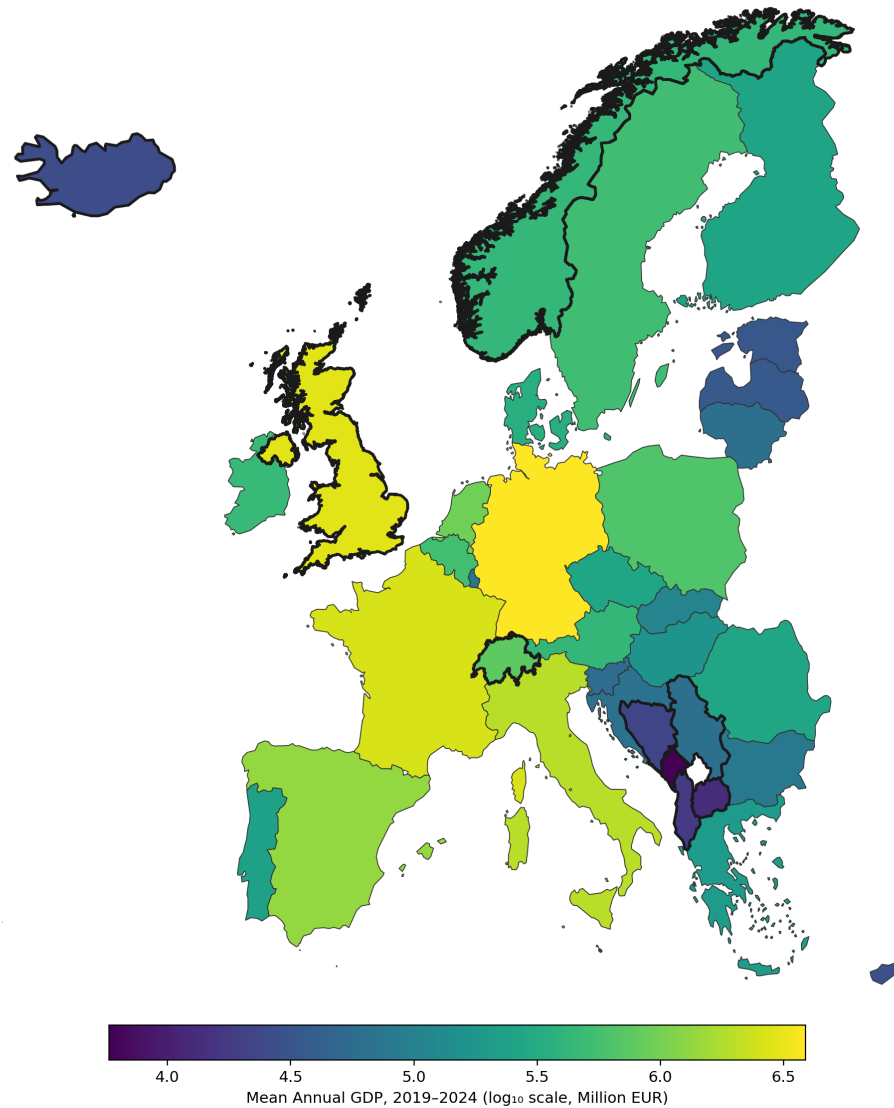
Green Policy Intelligence Engine (GPIE) - Source: CGLS NDVI 300m, Sentinel Hub Statistical API

GDP Choropleth Map

Control variable - GDP by country (annual, repeated across months in the panel).

GDP: EU-27 vs. 9-Country Control Group

Thick borders mark non-EU control-group countries — GDP used as an economic control variable in the causal model



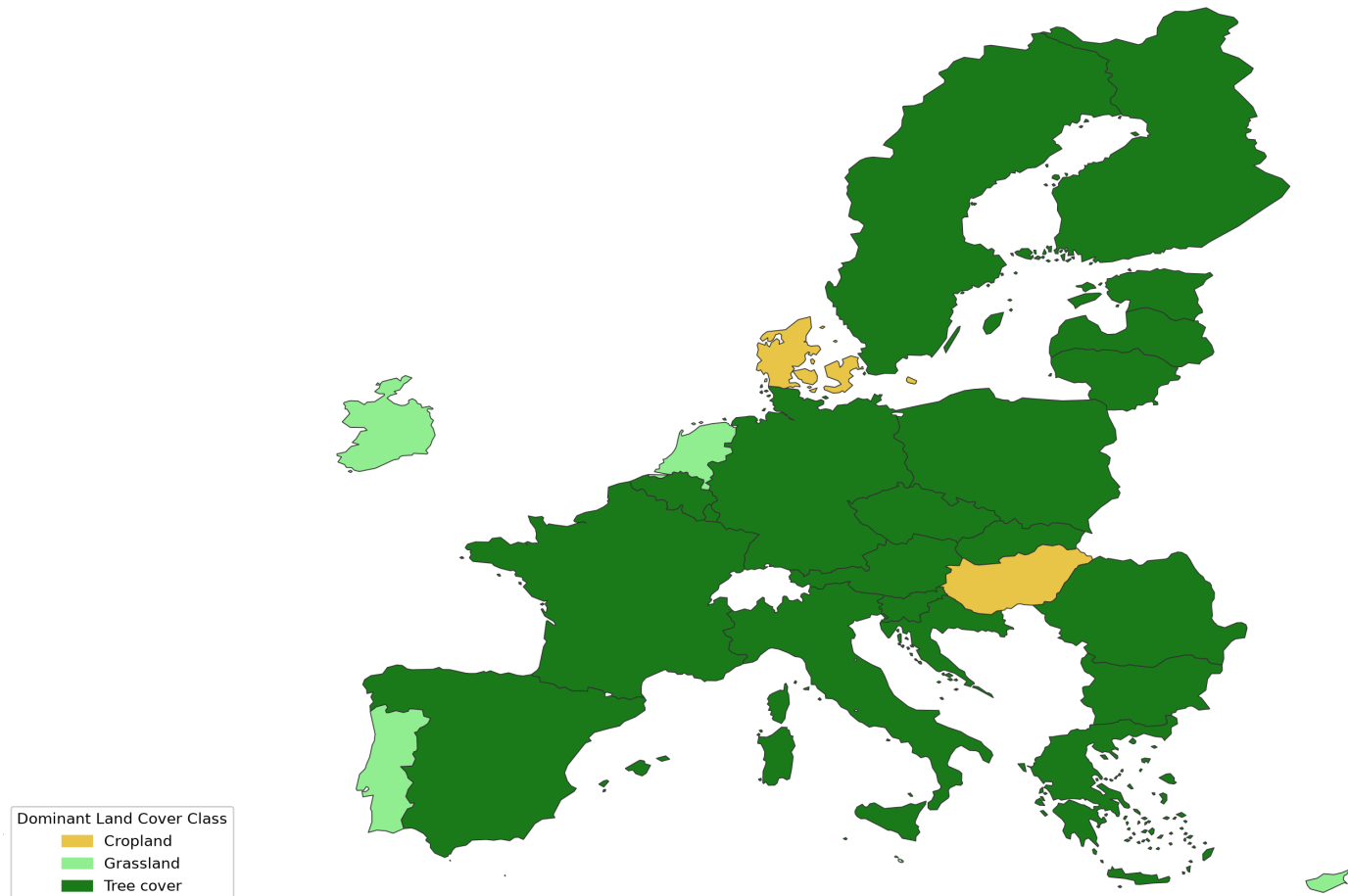
Green Policy Intelligence Engine (GPiE) — Source: Eurostat (EU-27), World Bank (control group)

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Dominant Land Cover Class Map

Descriptive context - dominant land-cover class (static; absorbed by country fixed effects, not a model regressor).

Dominant Land Cover Class by Country (EU-27)
Each country shaded by its single largest ESA WorldCover class (2021)



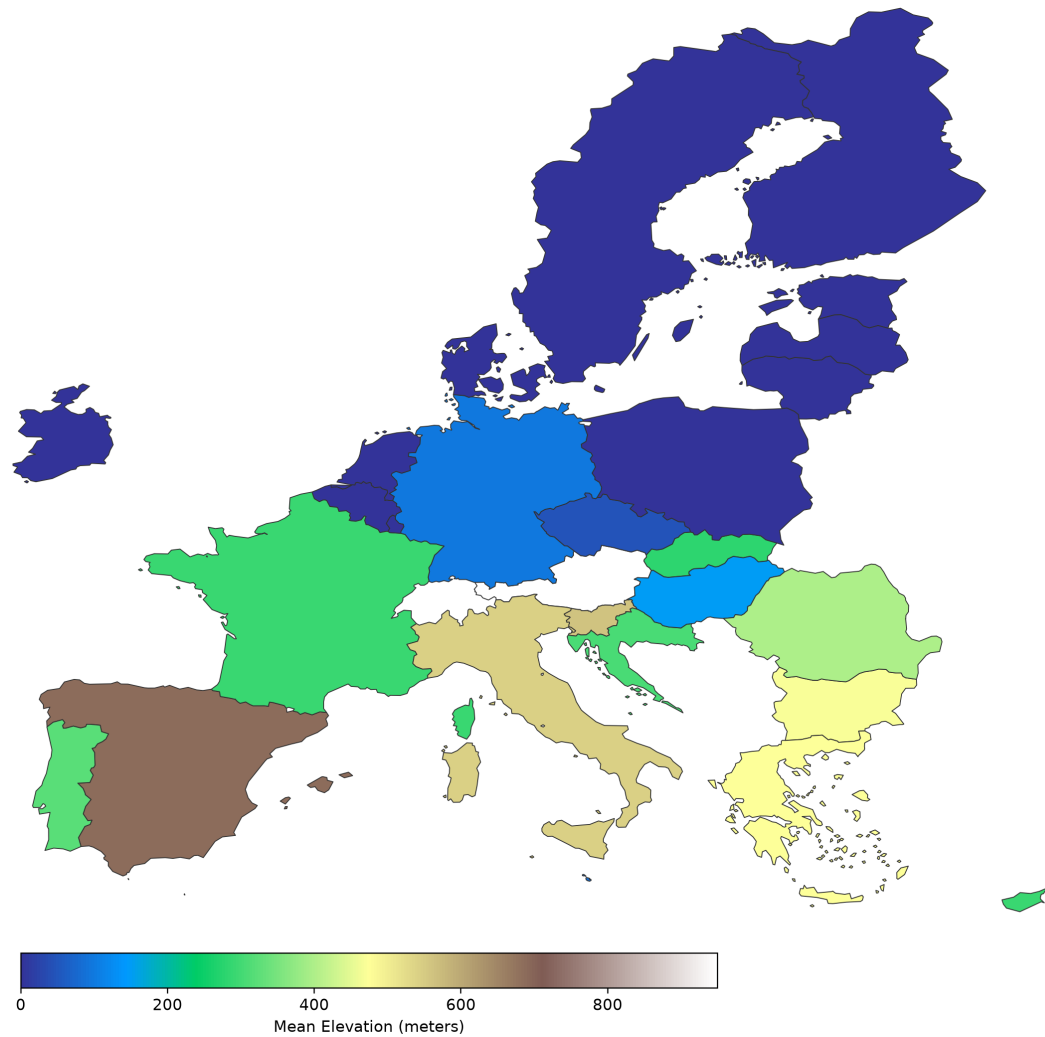
Green Policy Intelligence Engine (GPIE) — Source: ESA WorldCover 10m v200

GPIE - Green Policy Intelligence Engine

Elevation (DEM) Map

Descriptive context - mean elevation (static; absorbed by country fixed effects, not a model regressor).

Mean Elevation by Country (EU-27)
Environmental/topographic context — Copernicus DEM GLO-30

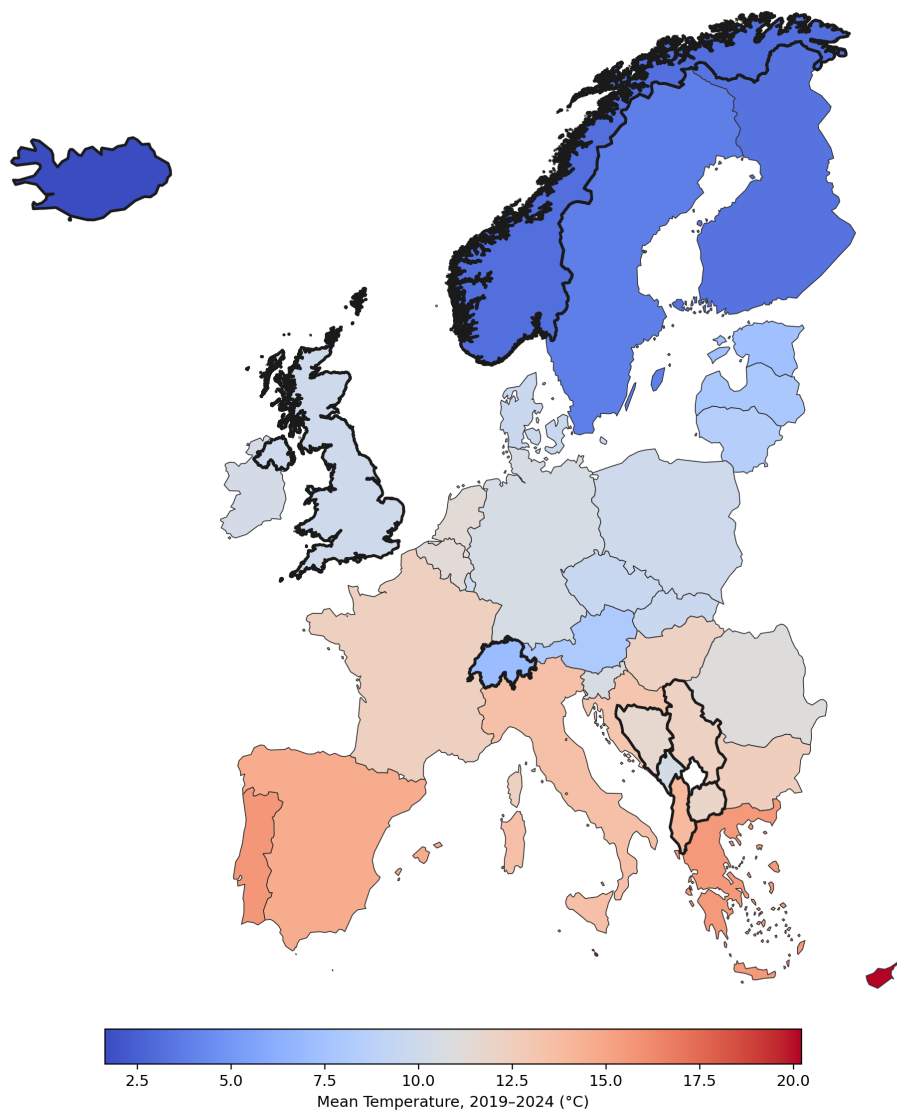


Climate / Temperature Map

Control variable - average temperature by country.

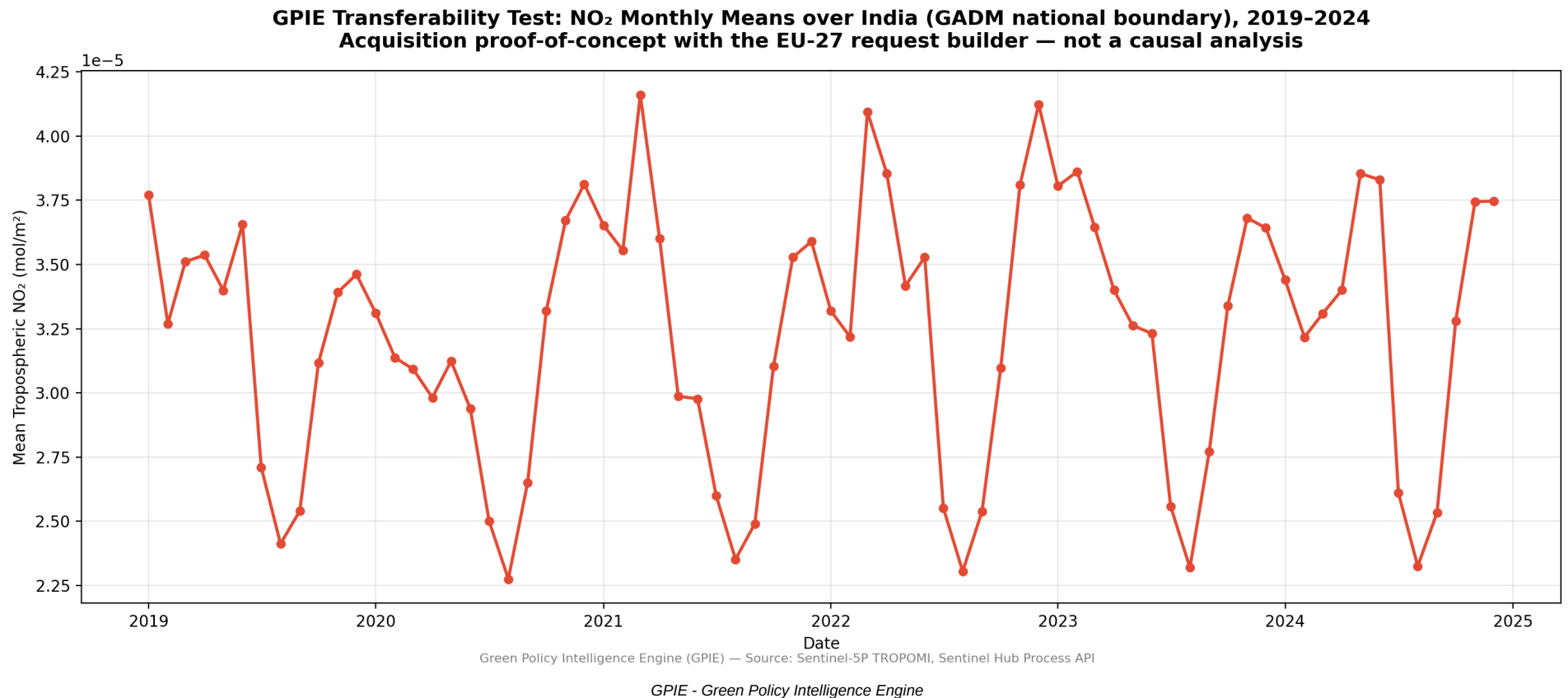
Mean Temperature: EU-27 vs. 9-Country Control Group

Thick borders mark non-EU control-group countries — temperature used as a climate control variable in the causal model



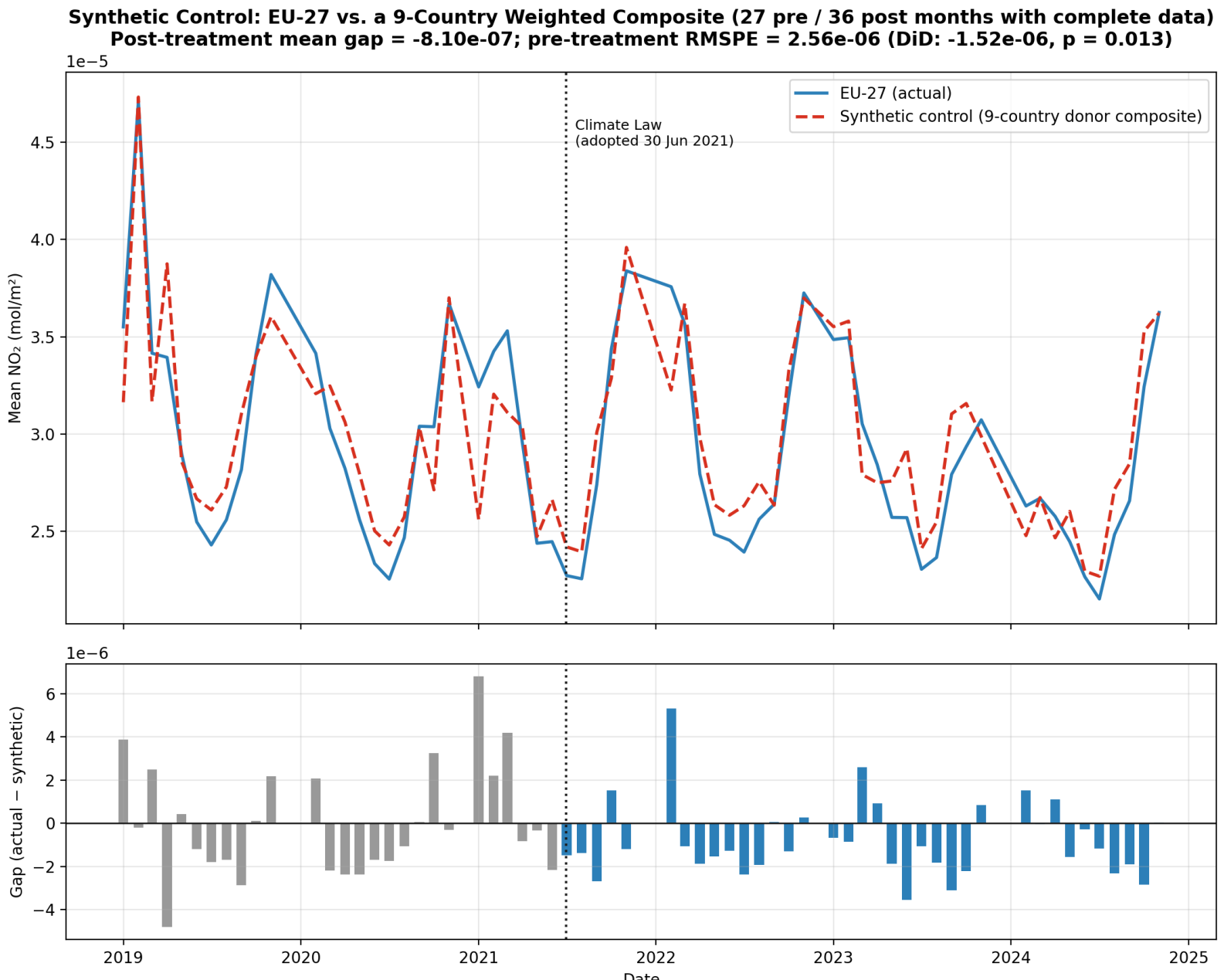
India Transferability Test

NO₂ monthly means over India's national boundary (GADM 4.1), 2019-2024, using the EU-27 request builder. Acquisition test only, not a causal analysis.



Synthetic Control: EU-27 vs. 9-Country Donor Composite

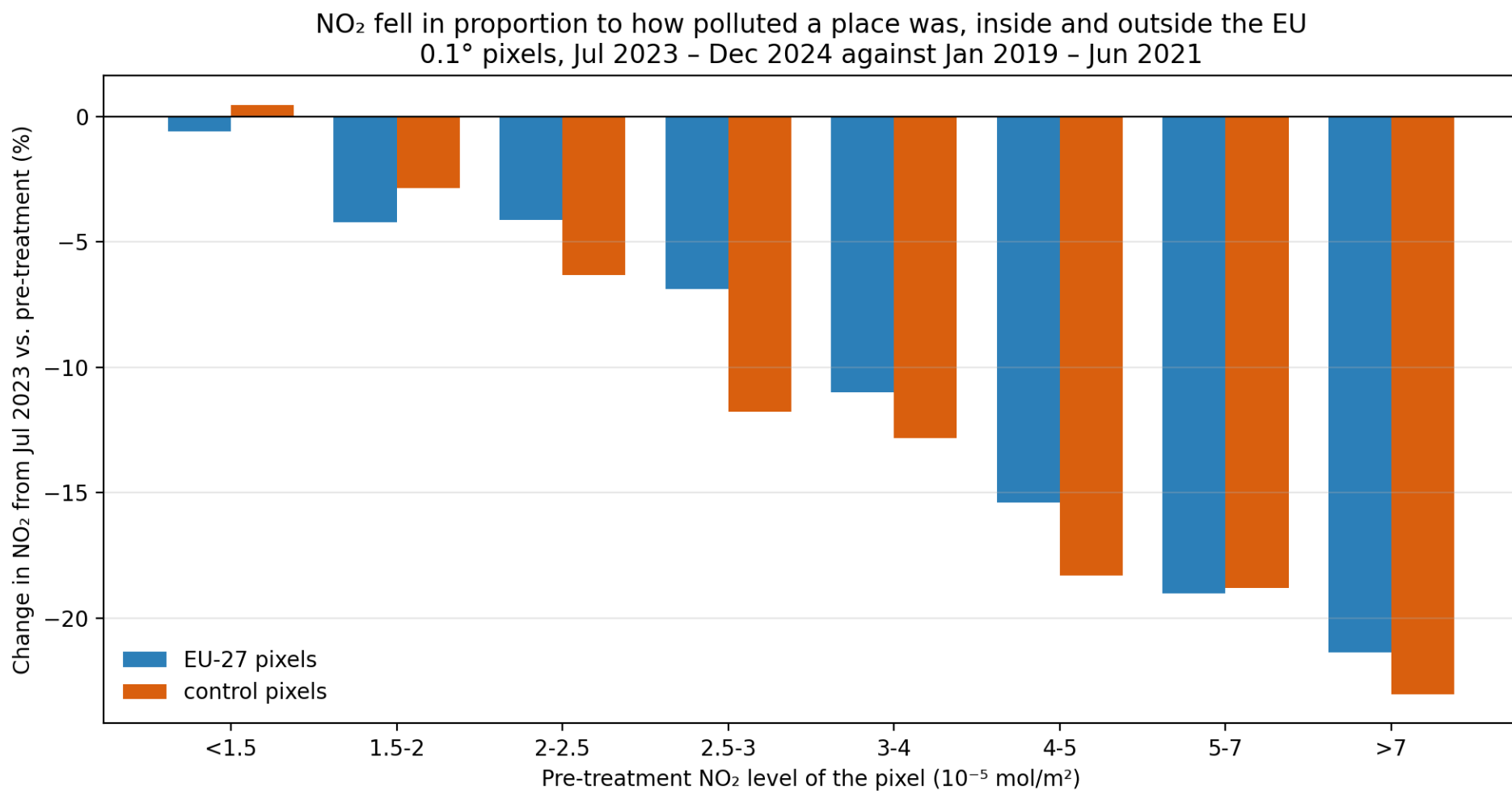
Convex donor weights with pre-period intercept adjustment; complete-data months only. Gap, pre-period RMSPE and DiD estimate in the figure title. All nine comparison countries are donors.



Green Policy Intelligence Engine (GPIE) — synthetic control with intercept adjustment; all nine comparison countries as donors; complete-data months only

NO₂ Change by Pre-Treatment Pollution Level

Change from July 2023 against the pre-treatment period, by the pixel's pre-treatment NO₂ level, for EU-27 and comparison-group pixels (0.1 degree grid).

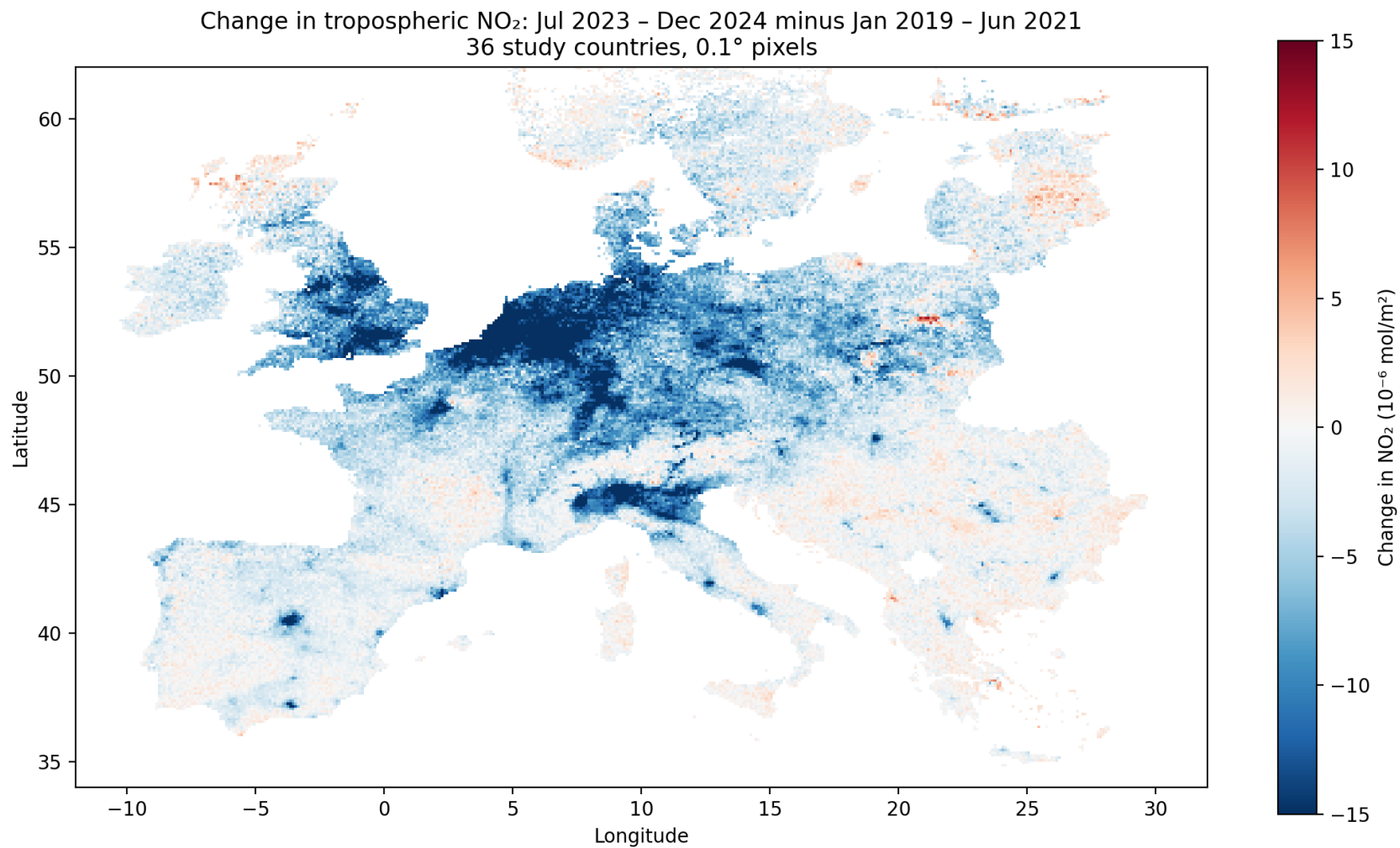


Green Policy Intelligence Engine (GPIE) - Source: Sentinel-5P TROPOMI, Sentinel Hub Process API

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Pixel-Level NO2 Change Map

Change in tropospheric NO₂ per 0.1 degree pixel, July 2023 - December 2024 minus January 2019 - June 2021.



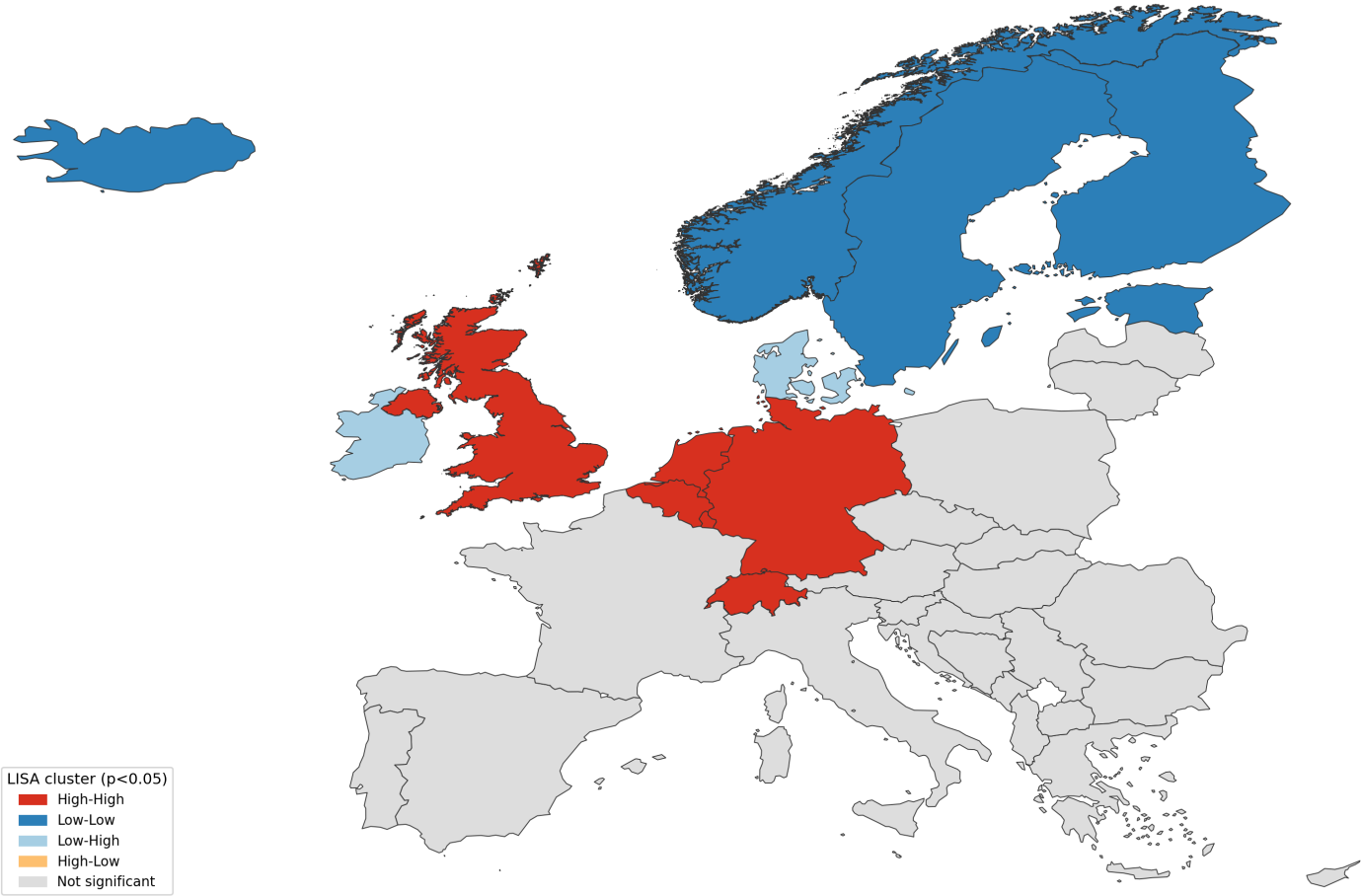
Green Policy Intelligence Engine (GPIE) - Source: Sentinel-5P TROPOMI, Sentinel Hub Process API

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Local Moran's I (LISA) Spatial Cluster Map

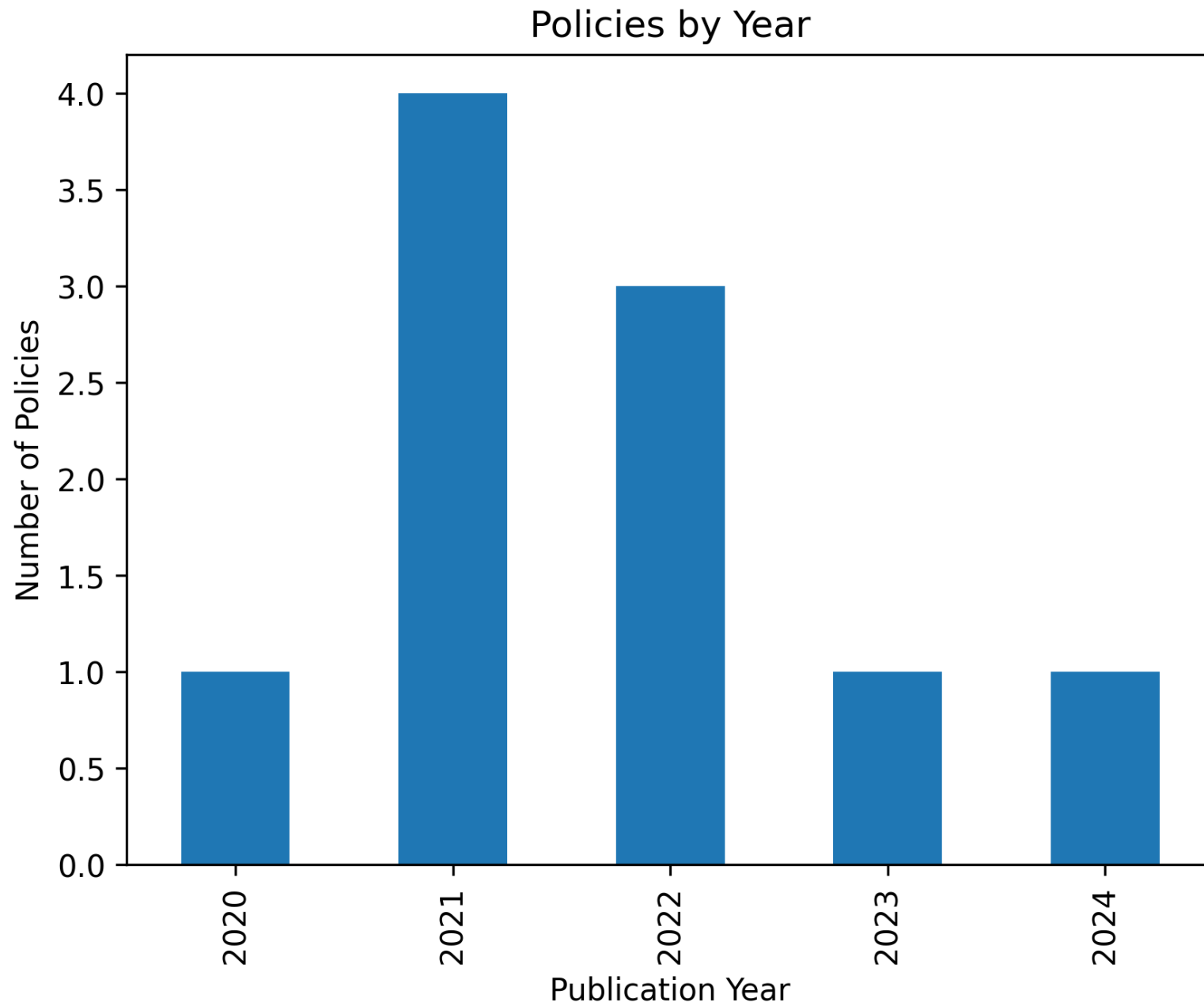
LISA clusters of full-period average NO2 (descriptive). Global and month-by-month residual Moran's I results in the figure title.

Local Moran's I: Where NO₂ Levels Cluster Spatially (Full-Period Average, 36 Countries)
Global Moran's I on average NO₂ level = 0.578 (p = 0.001);
DiD residuals, month by month: median I = 0.347, significantly positive in 59 of 72 months



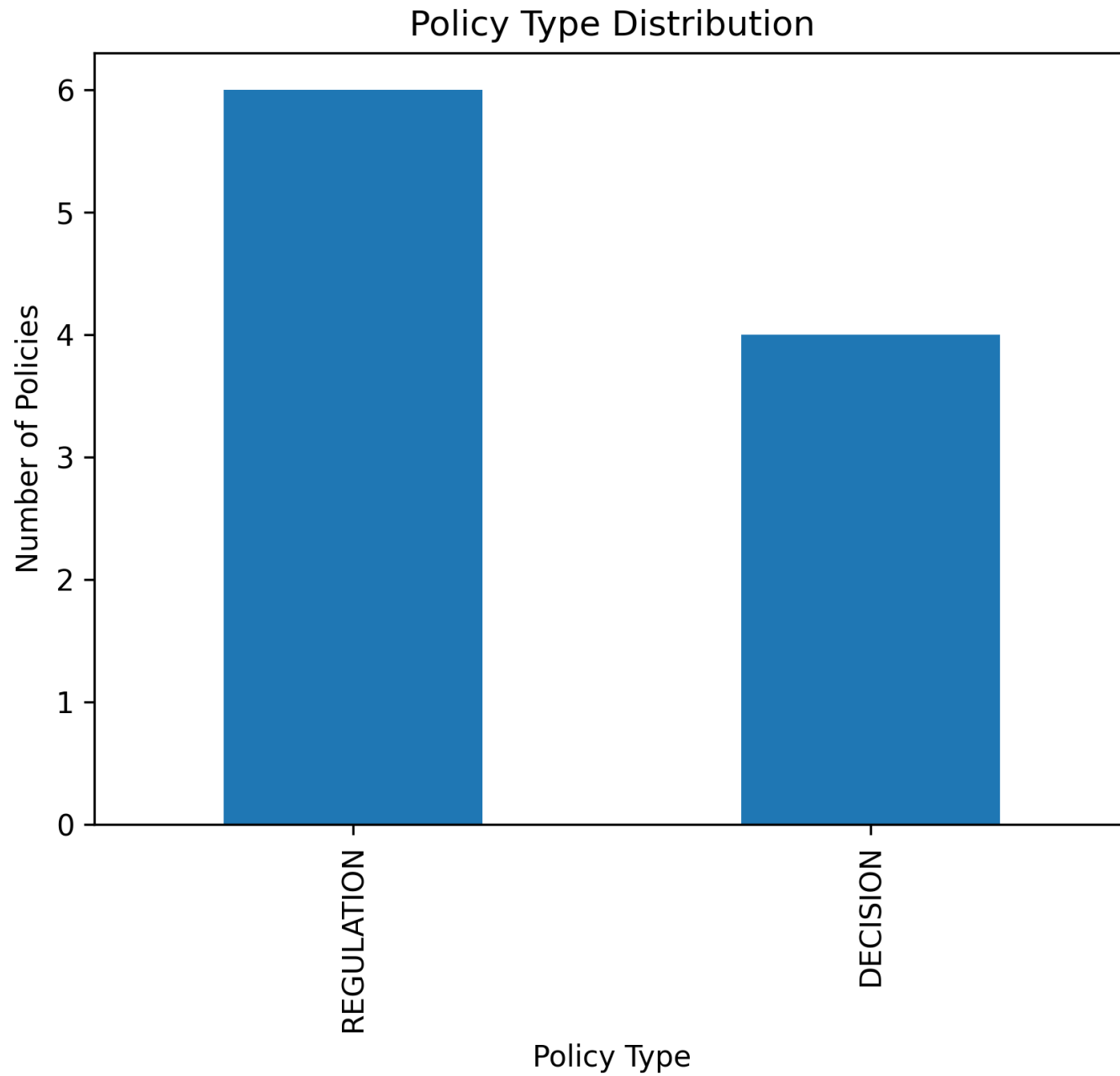
Policies by Year

Count of records per year in the 10-record EUR-Lex sample scraped for this project (not a complete policy inventory).



Policy Type Distribution

Breakdown of policy dataset by policy type.



Policy Types by Year

Policy type composition over time.

